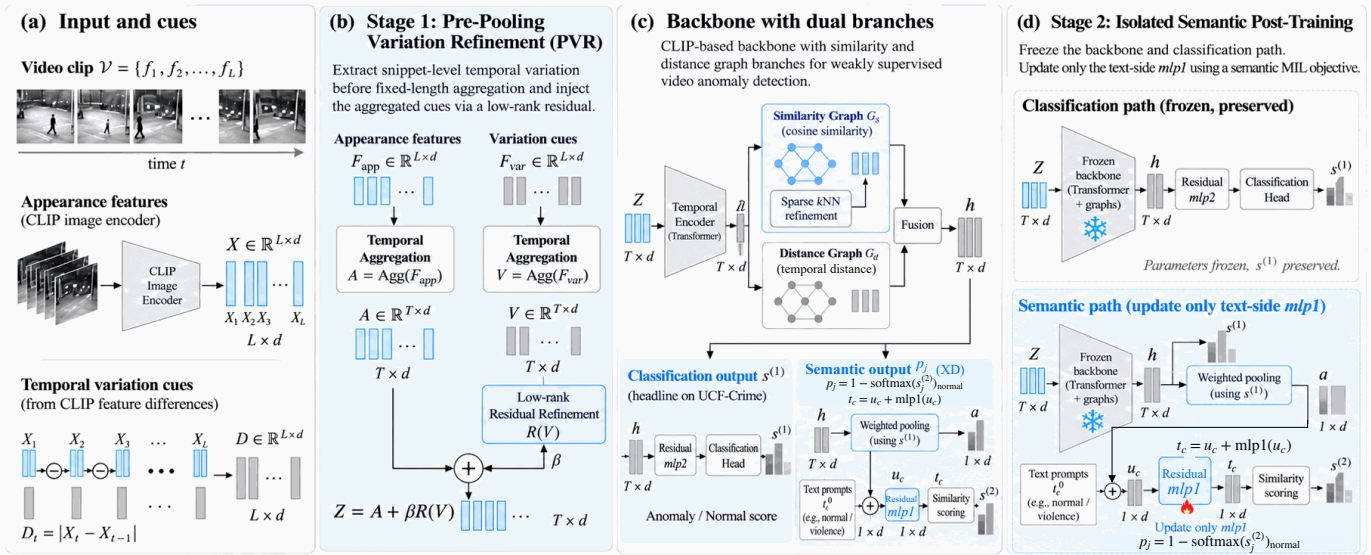

(a) Input and cues
Video clip $\mathcal{V} = \{f_1, f_2, \ldots, f_L\}$
time t
Appearance features
(CLIP image encoder)
CLIP Image Encoder
$X \in \mathbb{R}^{L \times d}$
X_1 X_2 X_3 X_L
$L \times d$
Temporal variation cues
(from CLIP feature differences)
X_1 X_2 X_3 X_t $D \in \mathbb{R}^{L \times d}$
$D_t = |X_t - X_{t-1}|$
$L \times d$
(b) Stage 1: Pre-Pooling Variation Refinement (PVR)
Extract snippet-level temporal variation before fixed-length aggregation and inject the aggregated cues via a low-rank residual.
Appearance features
$F_{\text{app}} \in \mathbb{R}^{L \times d}$
Variation cues
$F_{\text{var}} \in \mathbb{R}^{L \times d}$
Temporal Aggregation
$A = \text{Agg}(F_{\text{app}})$
Temporal Aggregation
$V = \text{Agg}(F_{\text{var}})$
$A \in \mathbb{R}^{T \times d}$
$V \in \mathbb{R}^{T \times d}$
$T \times d$
$T \times d$
Low-rank Residual Refinement
$R(V)$
β
$Z = A + \beta R(V)$
$T \times d$
(c) Backbone with dual branches
CLIP-based backbone with similarity and distance graph branches for weakly supervised video anomaly detection.
Z
$T \times d$
Temporal Encoder (Transformer)
$\tilde{h}$
$T \times d$
Similarity Graph G_s
(cosine similarity)
Sparse kNN refinement
Distance Graph G_d
(temporal distance)
Fusion
h
$T \times d$
Classification output $s^{(1)}$
(headline on UCF-Crime)
h
$T \times d$
Residual mlp2
Classification Head
$s^{(1)}$
Anomaly / Normal score
Semantic output p_j (XD)
$p_j = 1 - \text{softmax}(s_j^{(2)})_{\text{normal}}$
$t_c = u_c + \text{mlp1}(u_c)$
h
$T \times d$
Weighted pooling (using $s^{(1)}$)
a
$1 \times d$
Text prompts t_c^0 (e.g., normal / violence)
$1 \times d$
u_c
t_c
Residual mlp1
$1 \times d$ $1 \times d$
Similarity scoring
$s^{(2)}$
(d) Stage 2: Isolated Semantic Post-Training
Freeze the backbone and classification path.
Update only the text-side mlp1 using a semantic MIL objective.
Classification path (frozen, preserved)
Z
$T \times d$
Frozen backbone (Transformer + graphs)
h
$T \times d$
Residual mlp2
Classification Head
$s^{(1)}$
Parameters frozen, $s^{(1)}$ preserved.
Semantic path (update only text-side mlp1)
Z
$T \times d$
Frozen backbone (Transformer + graphs)
h
$T \times d$
$s^{(1)}$
Weighted pooling (using $s^{(1)}$)
a
$1 \times d$
$t_c = u_c + \text{mlp1}(u_c)$
Text prompts t_c^0 (e.g., normal / violence)
$1 \times d$
u_c
t_c
Residual mlp1
$1 \times d$ $1 \times d$
Similarity scoring
$s^{(2)}$
Update only mlp1
$p_j = 1 - \text{softmax}(s_j^{(2)})_{\text{normal}}$